```
import math

board = ['' for _ in range(9)]

def print_board():
  for i in range(3):
    print(board[3*i],'|',board[3*i+1],'|',board[3*i+2])
    if i<2:
      print("---------")

def is_winner(b,player):
  wins=[
      (0,1,2),(3,4,5),(6,7,8),
      (0,3,6),(1,4,7),(2,5,8),
      (0,4,8),(2,4,6)
  ]
  for a,b_,c in wins:
    if b[a]==b[b_]==b[c]==player:
      return True
  return False

def is_board_full(b):
  return '' not in b

def minmax_alpha_beta(b,depth,alpha,beta,is_maximizing):
  if is_winner(b,'X'):
    return 10-depth
  if is_winner(b,'O'):
    return depth-10
  if is_board_full(b):
    return 0

  if is_maximizing:
    max_eval=-math.inf
    for i in range(9):
      if b[i]=='':
        b[i]='X'
        eval=minmax_alpha_beta(b,depth+1,alpha,beta,False)
        b[i]=''
        max_eval=max(max_eval,eval)
        alpha=max(alpha,eval)
        if beta<=alpha:
          break
    return max_eval
  else:
    min_eval=math.inf
    for i in range(9):
      if b[i]=='':
        b[i]='O'
        eval=minmax_alpha_beta(b,depth+1,alpha,beta,True)
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```
      b[i]=''
      min_eval=min(min_eval,eval)
      beta=min(beta,eval)
```



```
      beta=min(beta,eval)
      if beta<=alpha:
        break
  return min_eval

def best_move():
 best_score=-math.inf
 move=-1
 for i in range(9):
  if board[i]=='':
    board[i]='X'
    score=minmax_alpha_beta(board,0,-math.inf,math.inf,False)
    board[i]=''
    if score>best_score:
      best_score=score
      move=i
 return move

def play_game():
 print("board position are numbered 0 to 8:")
 print("0|1|2")
 print("-----------")
 print("3|4|5")
 print("-----------")
 print("6|7|8")
 print("-----------")
 print()

 while True:
  move= best_move()
  if move==-1:
    print("game over its a tie.")
    break
  board[move]='X'
  print("AI(X)plays move:",move)
  print_board()
  if is_winner(board,'X'):
    print("AI(X)wins")
    break
  if is_board_full(board):
    print("game over its a tie")
    break

  while True:
    try:
      human_move= int(input("enter your move(0-8):"))
```

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        if 0<=human_move<=8 and board[human_move]==':
          board[human_move]='O'
          break
        else:
          print("invalid move try again")
      except ValueError:
        print("invalid input try again")
        print_board()
if is_winner(board,'O'):
```



```
    if is_winner(board,'O'):
      print("human(O)wins")
      break
    if is_board_full(board):
      print("game over its a tie")
      break

if __name__=="__main__":
  play_game()
```

```
board position are numbered 0 to 8:
0|1|2
-----------
3|4|5
-----------
6|7|8
-----------

AI(X)plays move: 0
X | |
---------
 | |
---------
 | |
enter your move(0-8):2
AI(X)plays move: 3
X | | O
---------
X | |
---------
 | |
```

enter your move(0-8):

12:31